

ε NFA :-

The NFA which has a transition even for empty string ϵ is called as ϵ NFA.

ϵ NFA Has five tuple. $\rightarrow$

$$M = \{Q, \Sigma, \delta, F, q_0\}$$

Where,

$Q \rightarrow$ Set of all states.

$\Sigma \rightarrow$ Input alphabet.

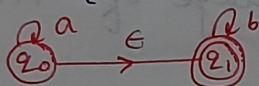
$\delta \rightarrow Q \times \Sigma \cup \epsilon \rightarrow 2^Q$ is a transition function.

$F \rightarrow$ set of final state.

$q_0 \rightarrow$ Initial state.

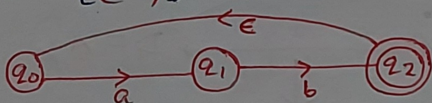
EX :-

1) $L = \{a^m b^n \mid m, n \geq 0\}$

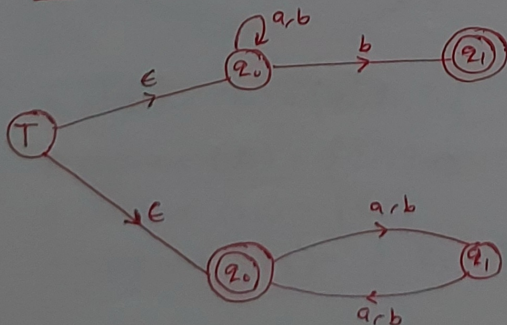
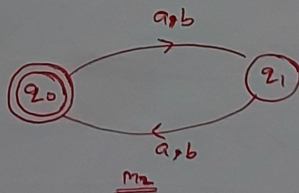
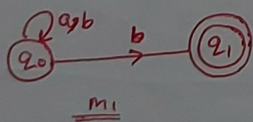


$$a^m \epsilon \cdot b^n = a \cdot b^n$$

2) $L = \{(ab)^n \mid n \geq 1\}$



EX :- Set of all string of a's and b's where every string ending with 'b' or the length is even.



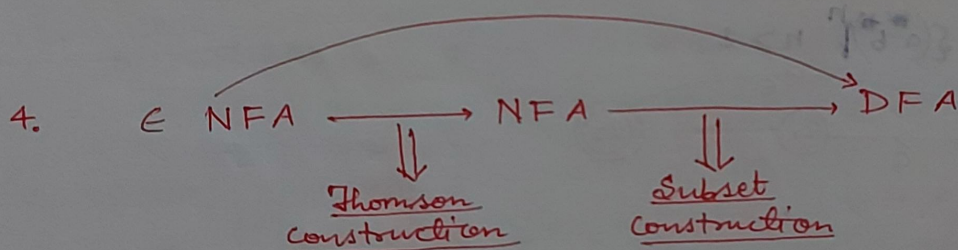
* Note :-

1. The Accepting power of ϵ NFA, NFA and DFA are same.

$$E(\epsilon\text{NFA}) = E(\text{NFA}) = E(\text{DFA})$$

2. Using ϵ NFA the system can move from current state to different state without ~~giving~~^{Taking} the input symbol from the input alphabet, Σ .

3. Inclusion or exclusion of ' ϵ ' transition will not affect the language of NFA.

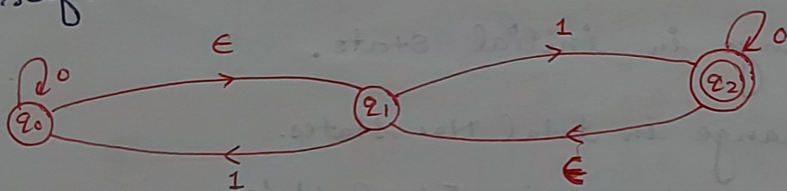


↓ ϵ closure (Q) :—

Let ' Q ' is any state in ENFA then the set of all state which are at zero distance from state ' Q ' is called as ϵ closure.

★ Note :—

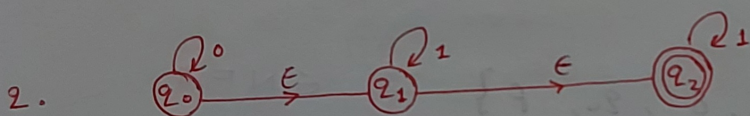
1. Every state is at zero distance from itself.



$$\epsilon \text{ closure } (q_0) = \{q_0, q_1\}$$

$$\epsilon \text{ closure } (q_1) = \{q_1\}$$

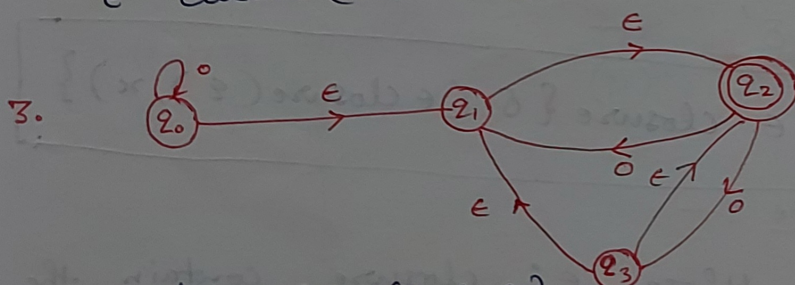
$$\epsilon \text{ closure } (q_2) = \{q_2, q_1\}$$



$$\epsilon \text{ closure } (q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon \text{ closure } (q_1) = \{q_1, q_2\}$$

$$\epsilon \text{ closure } (q_2) = \{q_2\}$$



$$\epsilon \text{ closure } (q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon \text{ closure } (q_1) = \{q_1, q_2\}$$

$$\epsilon \text{ closure } (q_2) = \{q_2\}$$

$$\epsilon \text{ closure } (q_3) = \{q_3, q_1, q_2\}$$

★ Note :-

$$\epsilon \text{ closure of } (\phi) = \phi$$

★ Conversion of ϵ NFA to NFA :-

1. No change in initial state.
2. No change in total No. states.
3. May be change in Final state.

★ Algorithm :-

$$M \Rightarrow \{Q, \Sigma, \delta, q_0, F\} \rightarrow \epsilon \text{ NFA}$$

$$M' \Rightarrow \{Q, \Sigma, \delta', q_0, F'\} \rightarrow \text{NFA}$$

★ Construction of δ' :-

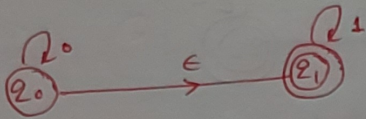
$$\delta'(q, x) = \epsilon \text{ closure } \{ \delta(\epsilon \text{ closure}(q), x) \}$$

★ For Final state :-

Every state whose ' ϵ ' closure contain the final state of ϵ NFA is the final state in NFA.

EX :-

1>



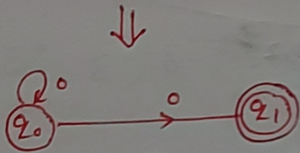
$$\epsilon \text{ closure}(q_0) = \{q_0, q_1\}$$

$$\epsilon \text{ closure}(q_1) = \{q_1\}$$

$$\begin{aligned} \delta'(q_0, 0) &= \epsilon \text{ closure} \{ \delta(\epsilon \text{ closure}(q_0), 0) \} \\ &= \epsilon \text{ closure} \{ \delta(q_0, 0), \delta(q_1, 0) \} \\ &= \epsilon \text{ closure} \{ q_0 \} \\ &= \boxed{q_0, q_1} \end{aligned}$$

$$\begin{aligned} \delta'(q_0, 1) &= \epsilon \text{ closure}(\delta(q_0, 1)) \\ &= \epsilon \text{ closure}(q_1) \\ &= q_1 \end{aligned}$$

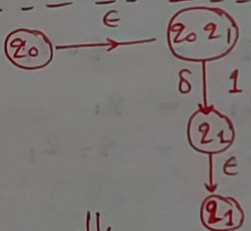
Ans



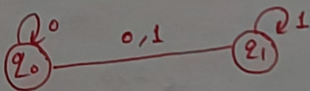
$$\delta'(q_1, 0) = \phi$$

$$\text{and } \delta'(q_1, 1) = q_1$$

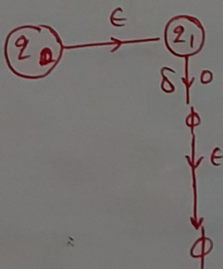
2>



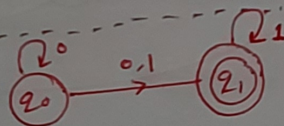
Ans



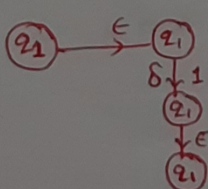
3>



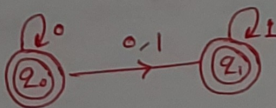
$\Rightarrow$



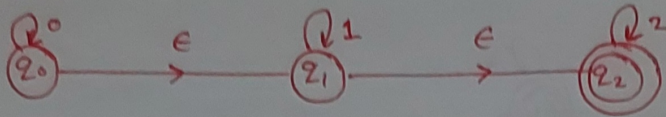
4>



$\Rightarrow$



Ex :-

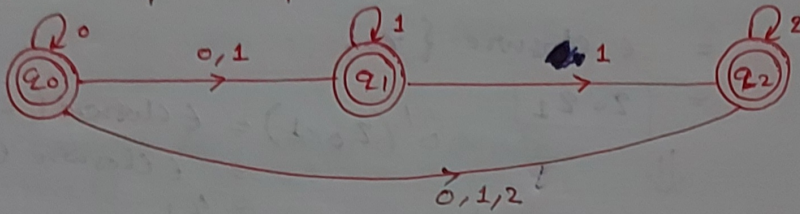


δ'	0	1	2
q_0	$q_0 q_1 q_2$	$q_1 q_2$	q_2
q_1	ϕ	$q_1 q_2$	q_2
q_2	ϕ	ϕ	q_2

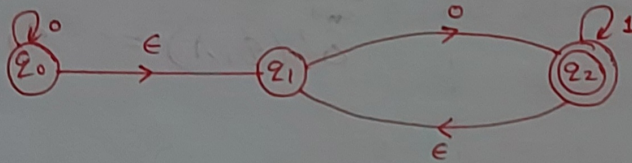
ϵ closure $q_0 = q_0 q_1 q_2$

ϵ closure $q_1 = q_1 q_2$

ϵ closure $q_2 = q_2$



Ex :-

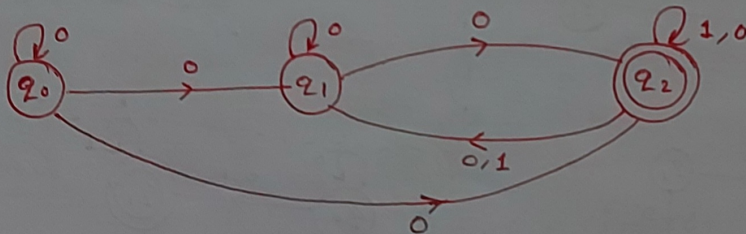


δ'	0	1
q_0	$q_0 q_1 q_2$	ϕ
q_1	$q_2 q_1$	ϕ
q_2	$q_1 q_2$	$q_2 q_1$

ϵ closure $q_0 = q_0 q_1$

ϵ closure $q_1 = q_1$

ϵ closure $q_2 = q_2 q_1$



Conversion of ϵ NFA to DFA

1. May be change in initial state.
2. May be change in total no. of state.
3. May be change in Final state.

* Algorithm :-

$$M = \{Q, \Sigma, \delta, q_0, F\} \longrightarrow \epsilon \text{ NFA}$$

$$M' = \{Q', \Sigma, \delta', q_0', F'\} \longrightarrow \text{DFA}$$

1. Initial State :-

$$q_0' = \epsilon \text{ closure}(q_0)$$

2. Construction of δ' :-

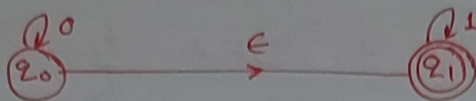
$$\forall q, x \quad \delta'(q, x) = \epsilon \text{ closure}\{\delta(q, x)\}$$

start the construction of δ' with the initial state and continue for every new state and stop the construction when ever no new state occur.

3. Final State :-

Every subset which contain the final state in ϵ NFA is final state in DFA.

Ex :-



$$\epsilon \text{ closure } (q_0) = \{q_0, q_1\}$$

$$\epsilon \text{ closure } (q_1) = \{q_1\}$$

step 1. Initial state $q_0' = \epsilon \text{ closure } (q_0)$

$$q_0' = q_0, q_1$$

q_0, q_1

$\delta \downarrow 0$

q_0

$\downarrow \epsilon$

q_0, q_1

step 2.

q_0, q_1

$\delta \downarrow 1$

q_1

$\downarrow \epsilon$

q_1

step 3 :-

q_1

$\delta \downarrow 0$

ϕ

$\downarrow \epsilon$

ϕ

Step 4 :-

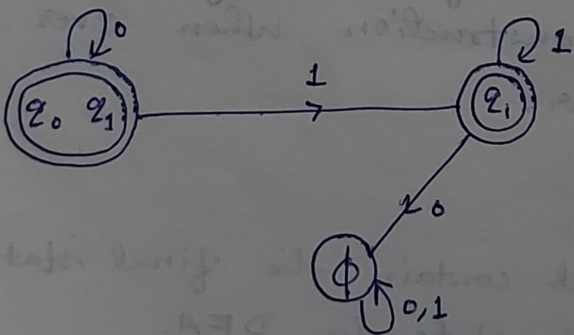
q_1

$\delta \downarrow 1$

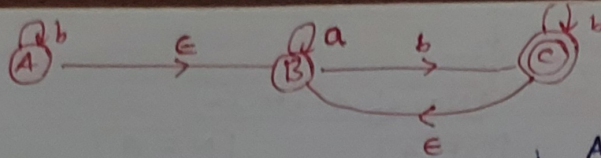
q_1

$\downarrow \epsilon$

q_1



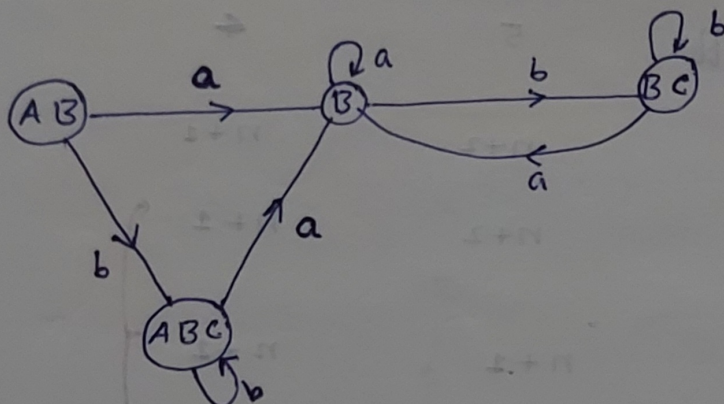
EX



ϵ closure of A = AB
 ϵ closure of B = B
 ϵ closure of C = CB

$AB \xrightarrow{a} B$
 $AB \xrightarrow{b} ABC$
 $ABC \xrightarrow{a} B$
 $ABC \xrightarrow{b} ABC$
 $BC \xrightarrow{a} B$
 $BC \xrightarrow{b} BC$

$B \xrightarrow{a} B$
 $B \xrightarrow{b} C$



Summary



No. of state in

S.N.

Language

DFA

NFA

1.

$L = \{w \in \Sigma^*, w = sx, |s| = m\}$

$m+2$

$m+1 \rightarrow$ From start

2.

$L = \{w \in \Sigma^*, w = xs, |s| = m\}$

$m+1$

$m+1 \rightarrow$ end From

3.

start and end with same symbol

5

4

4.

start and end with diff. symbol

5

4

5.

$|w|_a = n$

$n+2$

$n+1$

6.

$|w|_a \leq n$

$n+2$

$n+1$

7.

$|w|_a \geq n$

$n+1$

$n+1$

8.

$|w| = n$

$n+2$

$n+1$

9.

$|w| \leq n$

$n+2$

$n+1$

10.

$|w| \geq n$

$n+1$

$n+1$

11.

$|w|_a = r \pmod n$

n

n

12.

$|w| = r \pmod n$

n

n

13.

n^{th} symbol from left end is Fixed.

$n+2$

$n+1$

14.

n^{th} symbol from right end is Fixed

2^n

$n+1$

15.

$w = sx$ & $|w| = r \pmod n$, $|s| = m$

$m+n+1$

$m+n$

